Paleomagnetism of Upper Cretaceous Volcanic Rocks of Southwestern Montana

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Paleomagnetic studies of folded and flat-lying volcanic rocks in southwestern Montana indicate that most of the natural remanent magnetization (NRM) is primary and that secondary magnetization, where found, has been produced mainly by lightning. NRM directions of most rock units are grouped in either a normal or reversed sense after partial demagnetization, although directions of a few units of probable mudflow origin are highly dispersed. No evidence of self-reversed magnetization has been found; consequently, a field polarity sequence has been tentatively established in connection with Late Cretaceous and early Tertiary orogenic events. All folded rocks investigated belong to the Upper Cretaceous Elkhorn Mountains Volcanics; stratigraphic correlation between widely separated, structurally dissimilar outcrops of petrologically similar rocks is supported by NRM directions and polarities. Mean NRM directions of structurally restored, age-equivalent sites do not differ significantly from one another, which suggests that fold axes are nearly horizontal; however, they differ significantly from directions of nearby flat-lying volcanic rocks of probable Tertiary age. The virtual geomagnetic pole calculated for the Elkhorn Mountains Volcanics (69°N, 171°W) agrees generally with Late Cretaceous poles previously calculated by others for North America, Madagascar, and the U.S.S.R., but it differs significantly from Early Cretaceous poles of Japan and Australia.

INTRODUCTION

The remanent magnetization of Upper Cretaceous volcanic rocks in southwestern Montana was studied as an aid to interpreting the geologic history. Rocks suitable for paleomagnetic investigations were sampled from the foothills of Bull Mountain and other localities within a 12,500-km^2 area that is centered near 46°N, 112°W (Figure 1). Paleomagnetism of materials from Bull Mountain is of particular significance for two reasons—(1) the geologic age of the rocks there has been established and (2) the structural relations in the area are well known.

A paleomagnetic survey of extrusive rocks [*Colville*, 1961] indicated that several rock units near Bull Mountain possess a stable prefolding magnetization and that at least one rock unit has reversed remanent magnetization [*Colville and Mead*, 1960]. The present investigation is a detailed study of rocks near Bull Mountain, of other folded volcanic rocks believed to be part of the same volcanic field, and of nearby flat-lying younger extrusive rocks.

GEOLOGIC SETTING

All the extrusive rocks cropping out on the flanks and nearby foothills of Bull Mountain belong to the Elkhorn Mountains Volcanics described by *Klepper et al.* [1957]. The Elkhorn Mountains Volcanics, which crop out over an area of several thousand square kilometers, have been divided into three units that have an aggregate thickness of more than 3 km. The lower unit consists of fragmental rocks and autobrecciated lava flows; the middle unit consists of sheets of welded tuff interlayered with pyroclastic and epiclastic volcanic rocks and autobrecciated lava flows; and the upper unit consists of reworked volcanic rocks and pyroclastic rocks [*Klepper and Smedes*, 1959].

The volcanic rocks of Bull Mountain are part of a sequence some 2.9 km thick [*Alexander*, 1955] whose outcrops are nearly contiguous with those of the Elkhorn Mountains. These rocks probably are equivalents of the lower and middle units of the sequence studied in the Elkhorn Mountains.

The age of the Elkhorn Mountains Volcanics has been established as Late Cretaceous. The

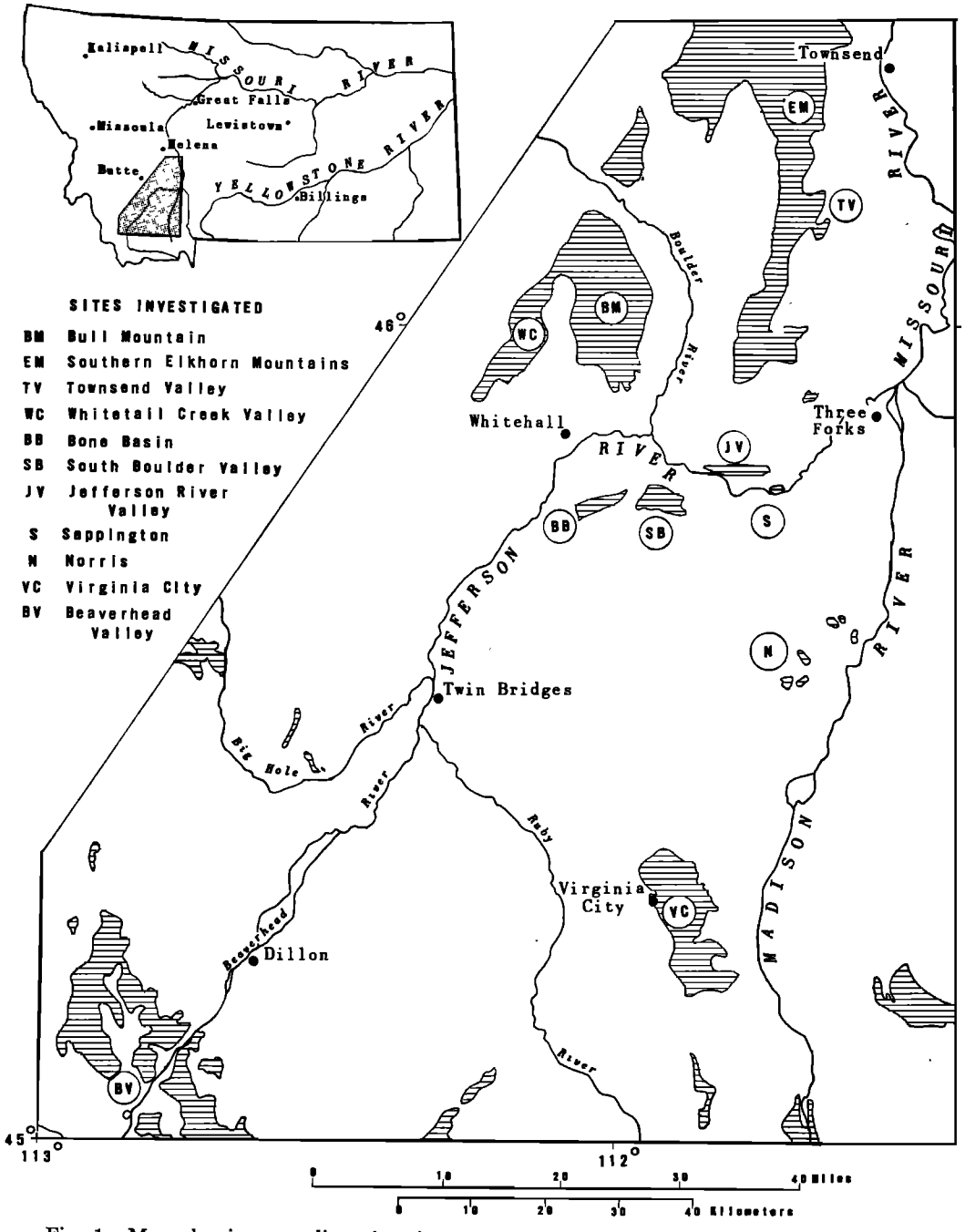


Fig. 1. Map showing sampling sites in southwestern Montana. Darkened areas indicate outcrops of volcanic rocks.

lowermost volcanic rocks overlie the Upper Cretaceous Slim Sam Formation; however, the original top of the volcanic pile may not be preserved. A variety of faunal and floral fossil remains from marly layers in conglomerate on the east flank of Bull Mountain have been assigned a probable Late Cretaceous age [*Klepper et al.*, 1957]. The conglomerate is believed

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to have been deposited near the end of the period of volcanism, suggesting that the volcanic rocks are wholly Cretaceous, probably middle Late Cretaceous [*Klepper et al.*, 1957]. Additional control on the upper age boundary is provided by age determinations of the Boulder batholith, a multiple intrusion which metamorphosed many volcanic rock units near Bull Mountain. Results of five lead-alpha age determinations on rocks from the Boulder batholith varied from 61 to 72 m.y. [*Chapman et al.*, 1955]; refined data gave 71 ± 4 m.y. as the mean age of the batholith [*Gottfried et al.*, 1959]. More recent results of eleven K-Ar and Rb-Sr age determinations from four different laboratories ranged from 68 to 82 m.y. [*Knopf*, 1964], which suggests that the batholith was emplaced late in the Cretaceous.

All folded rocks examined in the present investigation are located within 30 km of the Bull Mountain site, and all are probably part of the Elkhorn Mountains volcanic field. The outcrops are located in the southern Elkhorn Mountains described by *Klepper et al.* [1957], the Townsend Valley described by *Freeman et al.* [1958], the Whitetail Creek Valley, Bone Basin, the South Boulder Valley, and the Jefferson Valley near Lewis and Clark Cavern State Park (Figure 1). The rocks from these sites are generally similar to those of Bull Mountain, and stratigraphic and structural relations suggest that their ages are the same. A single inconsistency appears in Jefferson Valley, where leucite-rich basalt crops out among otherwise calc-alkalic rocks.

Flat-lying volcanic rocks and associated mafic intrusives were sampled at four sites, 25 to 120 km south of the Bull Mountain site. These sites are near Sappington, Norris, and Virginia City, and southwest of Dillon (Figure 1). All the rocks are believed to be Tertiary or younger. Basaltic intrusives at the Sappington site are post-Eocene and pre-Pleistocene [*Robinson*, 1963]; basaltic flows of the Norris and Virginia City sites are probably Pliocene or early Pleistocene [*Tansley et al.*, 1933]; and nearly flat-lying extrusives and associated intrusives southwest of Dillon are of two distinct ages—one probably Eocene and the other probably late Miocene or Pliocene [*Lowell*, 1965].

Analysis of Paleomagnetic Data

The NRM of cube-shaped rock specimens was measured with a spinner magnetometer belonging to the Indiana University Department of Geology. The limit of reliable sensitivity of the magnetometer was of the order of 5×10^{-6} emu/cm^3, and NRM directions were reproducible to within 1° or, exceptionally, 2°. The maximum error involved in the collection and preparation of oriented samples was estimated to be 3°.

To eliminate secondary magnetization that might give rise to a dispersion of the original NRM directions, rock specimens were partially demagnetized in alternating fields. Peak fields of 100 to 450 oe were generally sufficient to randomize the secondary magnetization. Data from rocks containing secondary components were analyzed statistically only after the specimens were partially demagnetized. A correction for tilt was applied to mean NRM directions associated with inclined volcanic strata.

Because of the difficulty of determining exact orientations of fold axes in the field, all tilt corrections were applied by rotation about the present geologic strike; that is, a horizontal fold axis is assumed. Where fold axes could be estimated from field relations, none plunged more than 5°.

The dispersion of NRM directions was studied by the methods of *Fisher* [1953]. Statistical methods were applied at three levels: (1) to mean directions of oriented samples within a rock unit, (2) to mean directions of rock units at a given site, and (3) to mean site directions of an entire formation. Directions of individual oriented samples were obtained by simply averaging the directions of two or three specimens from the sample. As an additional measure of dispersion, independent of the population density distribution, values of angular standard deviation [*Wilson*, 1959] were calculated.

Virtual geomagnetic poles (VGP) [*Cox and Doell*, 1960] or paleomagnetic poles (according to the usage of *Irving* [1964]) were calculated for most sites, mean rock unit directions being used as data points. For the few sites where only a single rock unit is exposed, mean sample directions were used in calculating the virtual pole. A VGP calculated for a single rock unit

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TABLE 1. NRM of Rock Units from the Bull Mountain Site after Partial Demagnetization of Samples

Rock Unit	N	D, deg	I, deg	α_{95}, deg	k	Polarity	$J_n \times 10^4$, emu/cm³	Peak Demagnetizing Field, oe	δ, deg	Remarks
O	7	100	−59	15	13	R	1.8	400	21	Ash flow tuff
N	19	88	−26	15	5	R	0.22	400	37	Welded tuff
M	32	79	−30	6	17	R	1.1	100	19	Welded tuff
L	10	82	−33	7	38	R	1.6	400	12	Welded tuff, oxidized part of unit M
K	8	265	53	9	30	N	2.6	100	14	Tuffaceous sandstone, baked
J	5	149	67			N	7.8	400	25	Andesitic intrusive rock; directions not axially symmetric; Fisher's statistics not applied
I	6	255	39	8	51	N	4.3	250	10	Basalt, baked
H	17	83	−60	14	6	R	7.9	100	33	Basalt
G	31	86	−32	5	33	R	4.8	450	14	Andesite flow
F	11	244	67	12	12	N	1.2	250	19	Pyroxene andesite, highly weathered; may be intrusive or extrusive in origin; probably CRM
E	12	71	−30	9	22	R	9.3	250	16	Tuff
D	5	249	54	7	80	N	0.96	400	8	Graywacke
C	10	290	30	6	56	N	99	400	10	Lithic tuff
B	5	260	58	11	33	N	0.61	100	12	Red graywacke
A	8	290	55	10	24	N	0.11	400	15	Tuffaceous siltstone

Note: Bedding correction has not been applied.
N, number of samples used in estimating mean direction.
D, declination of estimated mean direction, east of north.
I, inclination of estimated mean direction, below horizontal.
α_{95}, 95% confidence angle about the estimated mean direction.
k, between-sample precision parameter estimate.
Polarity, N if normal (same sense as present field); R if reversed.
J_n, intensity of NRM.
δ, angular standard deviation [*Wilson*, 1959].

represents the orientation of an hypothetical geocentric dipole at a particular instant of geological time. The oval of confidence [*Cox and Doell*, 1960] associated with such a pole is necessarily smaller than the oval associated with a pole representing the time-averaged ancient field.

Where both normal and reversed polarities occur, the polarity of least frequent occurrence was arbitrarily transposed to its antipodal position to determine the mean site direction. VGP's associated with reversed mean site directions were plotted as south poles for comparison with the south poles corresponding to normal mean site directions. Separate analyses of normal and reversed polarities at a given site have indicated that the antipodes of reversed polarities always lie within a few degrees of the normal polarities. Mixed polarities within a given rock unit were not encountered except in epiclastic materials whose depositional temperature was lower than the Curie points of the constituent magnetic minerals and in materials remagnetized by lightning.

Results

Bull Mountain site. Fifteen rock units were sampled from a 1070-km-thick volcanic sequence in the southeastern foothills of Bull Mountain. The volcanic pile here lies unconformably on Paleozoic limestones and a synorogenic conglomerate. Although exposures suitable for paleomagnetic sampling constitute only a small fraction of the total volcanic section, they consist of a variety of rock types including tuffs, flows, intrusives, and water-laid deposits. The results of measurements of NRM after partial demagnetization (cleaned NRM)

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are summarized in Table 1, and a cross section is shown in Figure 2.

The reliability of paleomagnetic data obtained from the Bull Mountain section is supported by the consistency of cleaned NRM directions, the consistency of reversed NRM directions which are grouped approximately 180° from the normal directions, and the grouping of all directions away from the present field direction. Results of partial demagnetization, together with considerations of terrain, suggest that seven rock units have been locally remagnetized by lightning. Samples containing considerable secondary magnetization, removable in peak alternating fields of 100 to 450 oe, characteristically occur in relatively open terrain that is presumably susceptible to lightning discharges. A detailed sampling of unit G indicated that remagnetized 'cells' [*Cox*, 1961] have an average areal extent of approximately 50 m². Original NRM directions of unit G are shown in Figure 3*a*; cleaned NRM directions are less highly dispersed, as shown in Figure 3*b*. NRM directions of other rock units affected by lightning show similar decreases in dispersion after partial demagnetization. Man-made exposures of unit M possess NRM directions typical of rocks in the section that have not been affected by lightning (Figure 4).

Magnetic data of units F and J, though indicating a stable original NRM, were not used in calculations of the virtual pole for the Bull Mountain site. Directions of unit F differ significantly from those of other normally magnetized rock units in the section. Extreme weathering of all exposures of this unit suggests that a chemical remanent magnetization (CRM) has been acquired after deposition or emplacement. NRM directions of unit J are anomalously distributed along a great circle as shown in Figure 5 and thus deviate from Fisher's distribution. NRM directions of units I and K, baked by the intrusion of unit J, are somewhat more symmetrically distributed (Figure 5).

The origin of reversed NRM polarities within the Bull Mountain section bears on the interpretation of Late Cretaceous geologic and magnetic histories. The possibility of instantaneous self-reversed magnetization was examined in heating and cooling experiments on specimens from each of the seven reversed

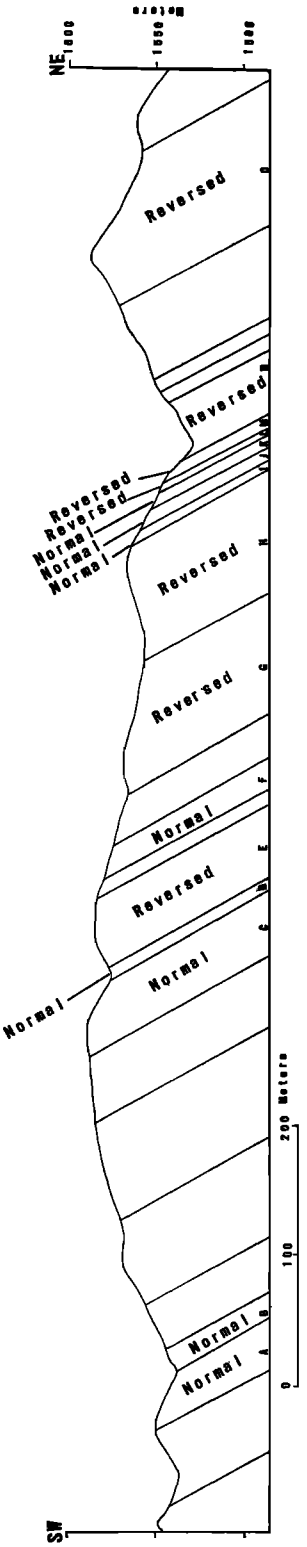


Fig. 2. Generalized cross section of the Bull Mountain site showing NRM polarities of the rock units sampled. Letters correspond to rock types given in Table 1.

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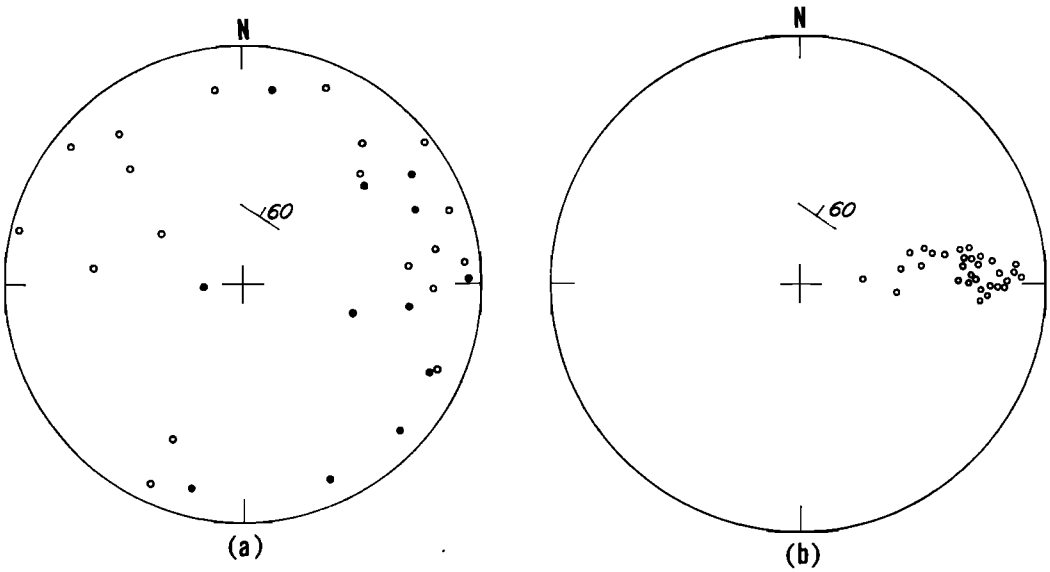


Fig. 3. (*a*) Original NRM directions of 31 samples from natural exposures of unit G, Bull Mountain site. (*b*) NRM directions of the samples after partial demagnetization in peak alternating fields of 450 oe. Open circles represent north-seeking directions on the upper hemisphere; closed circles represent directions on the lower hemisphere. Bedding is approximately N55°W, 60°NE. Schmidt equal-area projections.

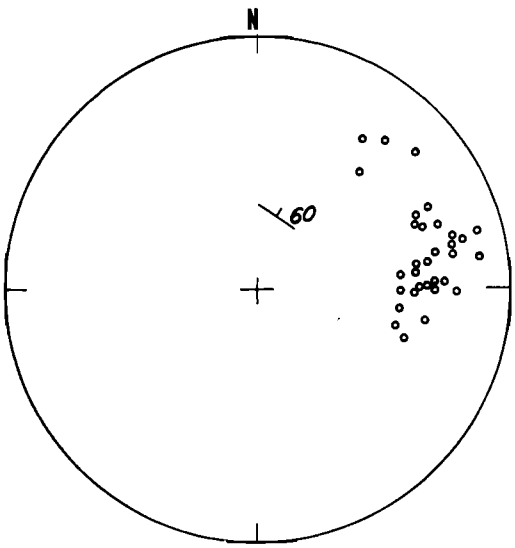


Fig. 4. Original NRM directions of 32 samples from man-made exposures of unit M, Bull Mountain site. Partial demagnetization in alternating fields did not appreciably change the original directions. Bedding is N55°W, 60°NE. Schmidt equal-area projection.

units sampled. In no case was the polarity of the acquired thermoremanent magnetization (TRM) opposite to that of the applied field. Petrographic examinations of thin sections, together with X-ray diffraction analyses of magnetic mineral separates, indicated that only one reversed unit contains a member of the ilmenite-hematite series. This unit contains titanomagnetite as its principal magnetic mineral constituent, although hematite occurs in oxidized rims of the magnetite grains. Chemical analyses of magnetic mineral separates of units G, H, L, and M indicate an abundance of titanium within the materials, as shown in the ternary diagram of Figure 6. The general absence of ilmenohematite appears to rule out the self-reversal mechanisms described by *Balsley and Buddington* [1954], *Uyeda* [1958], and *Ishikawa and Syono* [1963]. The possibilities of self-reversal mechanisms involving the cooling of spinel sublattices with appropriate temperature coefficients [*Gorter and Schulkes,* 1953] and cation migration between adjacent lattice sites [*Verhoogen,* 1956] cannot be dismissed on experimental evidence.

The alternative possibility of field reversals is supported by the agreement of polarities

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within the baked parts of units I and K, which are opposite to those of the volcanic section intruded by unit J. Field reversals are also suggested by general negative evidence for self-reversals; such evidence includes the diversity of rock types exhibiting reversed polarities and the total lack of mineralogic correlation between normal and reversed units. Reversed polarities occur in five tuffs and two flows, all of highly variable composition, whereas normal polarities occur in three waterlaid deposits, one tuff, one flow, and one intrusive with its associated baked zone. All units within the section contain cubic magnetite; three reversed and two normal units contain small quantities of cubic maghemite; and

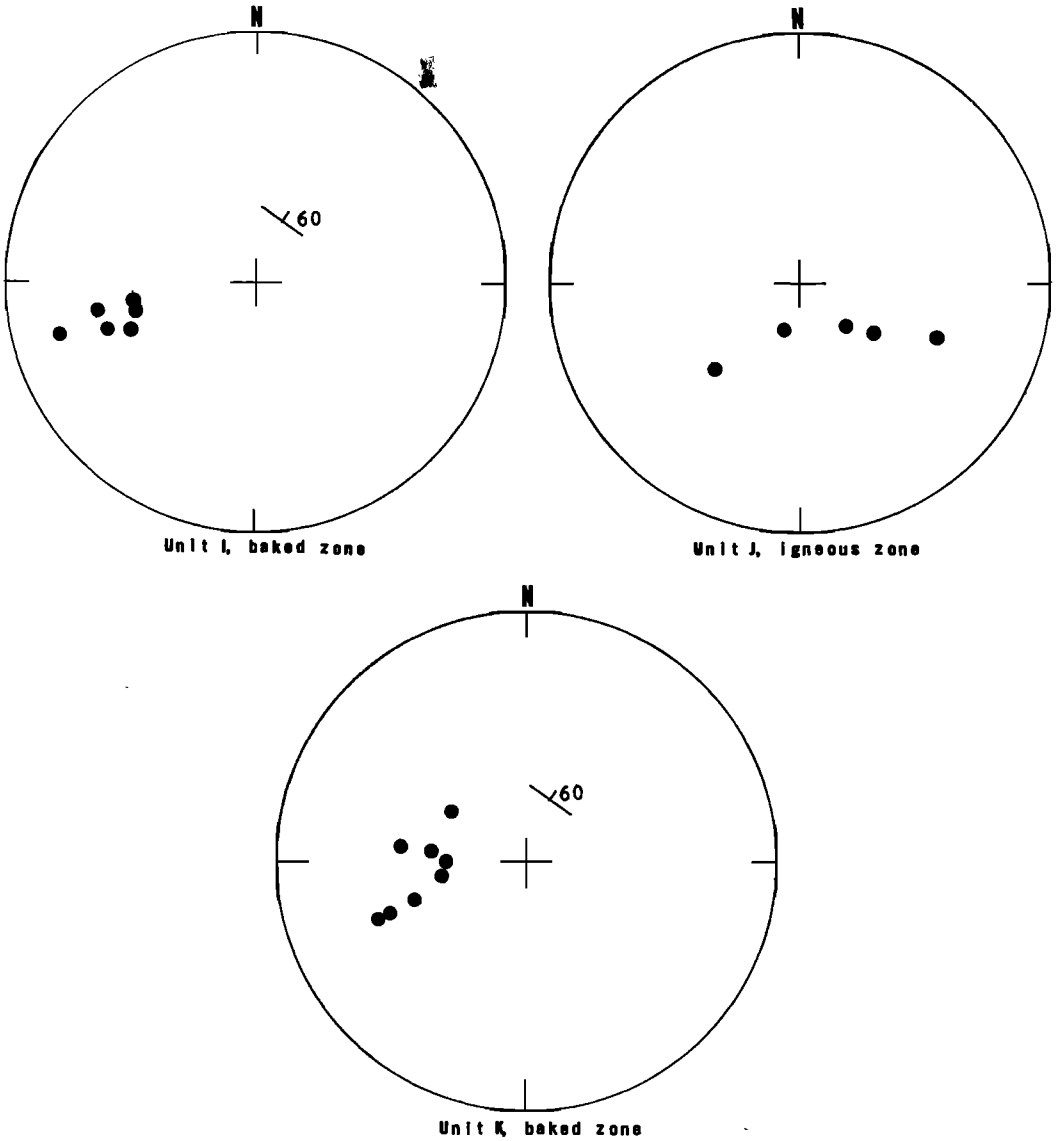


Fig. 5. NRM directions of unit J, an andesitic intrusive, and units I and K, baked zones stratigraphically below and above the intrusive, Bull Mountain site. Bedding of I and K is N55°W, 60°NE. Intrusive J is nearly conformable with bedding. The NRM of all units is stable in demagnetizing fields up to 450 oe. Schmidt equal-area projection.

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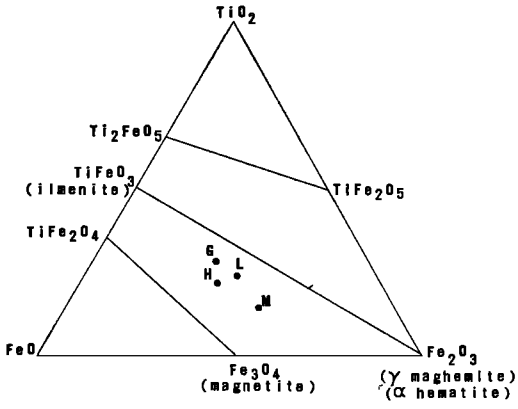


Fig. 6. Ternary system FeO-Fe_2O_3-TiO_2 showing compositions of magnetic mineral separates of units G, H, L, and M, Bull Mountain site.

one reversed and two normal units contain variable amounts of rhombohedral hematite.

The stratigraphic sequence of magnetic polarities indicated in Table 1 and Figure 2 might suggest a minimum of three normal-reversed cycles if the time sequence of magnetization were consistent with the stratigraphic sequence observed. However, as previously noted, the NRM of unit F is probably CRM acquired after folding and the NRM of units I and K is probably TRM acquired during the intrusion of unit J. These interpretations reduce the complexity of the polarity sequence to a minimum of one normal-reversed cycle during extrusion of the materials, followed by a normal half-cycle during emplacement of the intrusive. The simplest sequence of events which can be tentatively inferred is (1) deposition of units A, B, C, and D in a normal magnetic field; (2) deposition of units E, F, G, H, I, K, L, M, N, and O in a reversed magnetic field; (3) intrusion of unit J accompanied by baking of units I and K in a normal magnetic field before the main phase of folding; and (4) chemical alteration and partial remagnetization of unit F in a normal magnetic field after the main phase of folding. A more complete reconstruction of the polarity sequence awaits paleomagnetic sampling of materials not now well exposed within the section.

Mean NRM directions of rock units from the Bull Mountain site are listed in Table 1; a single tilt correction was applied to all directions given in Table 1 to obtain the estimated mean site direction before folding. The tilt correction applied is subject to error because the fold axis orientation could not be precisely determined and because possible initial dips of extrusive units were not specifically taken into account. Most units were sampled over an area sufficiently large to average unrecognized initial dips effectively. The uncertainty of fold axis orientation, however, remains a source of error in the determination of the VGP. Although this error is undoubtedly small at the Bull Mountain site, larger errors may result at sites where the fold axis orientation is less precisely known. Tilt-corrected mean directions of the sites of folded rocks are shown in Figure 7.

Other sites of folded volcanic rocks. Six other sites were investigated, and a brief description of each is given with a summary of the magnetic data obtained.

1. *Southern Elkhorn Mountains.* Three tuffs and one flow were sampled from exposures near high peaks approximately 30 km north-northeast of the Bull Mountain site. Parts of these extrusives have been thermally metamorphosed by the nearby Boulder batholith. All the rock units have been partially remagnetized by lightning, and partial demagnetization results indicate that one of the tuffs is reversely magnetized and the other units are normally magnetized. Directions within units remain somewhat highly dispersed even after application of demagnetizing fields. Mean directions of all units sampled were used to determine the VGP. Bedding is generally obscure.

2. *Townsend Valley.* Four tuffs, two flow breccias, and one hypabyssal intrusive were sampled on Lone Mountain, approximately 30 km northeast of the Bull Mountain site. All the extrusives have been thermally metamorphosed to some extent by the Lone Mountain stock, a monzonitic intrusive [*Freeman et al.*, 1958]. Parts of five units sampled have been remagnetized by lightning; partial demagnetization results indicate that all units are normally magnetized. At least two units have a probable mudflow origin; highly dispersed, cleaned NRM directions of these units suggest that the magnetic mineral constituents have been randomly deposited at temperatures below their Curie points, a problem discussed

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by *Aramaki and Akimoto* [1957]. Mean NRM directions of five units were used in the determination of the VGP. Bedding throughout the area is generally obscure.

3. *Whitetail Creek Valley.* Three tuffs and two autobrecciated intrusives were sampled from exposures near the contact of the Boulder batholith, approximately 8 km west of the Bull Mountain site. Two units have been remagnetized by lightning; partial demagnetization results indicate that all units are normally magnetized. All units were used for calculating the VGP. Bedding is generally obscure, though fairly well defined at a few exposures.

4. *Bone Basin.* Three tuffs and one flow were sampled from low-lying ridges near Bone Basin, approximately 15 km south-southwest of the Bull Mountain site. These extrusives are petrologically similar to units in the reversed part of the Bull Mountain sequence. Samples of all units possess secondary magnetization believed to be due to lightning discharges. Partial demagnetization results indicate that all units are reversely magnetized.

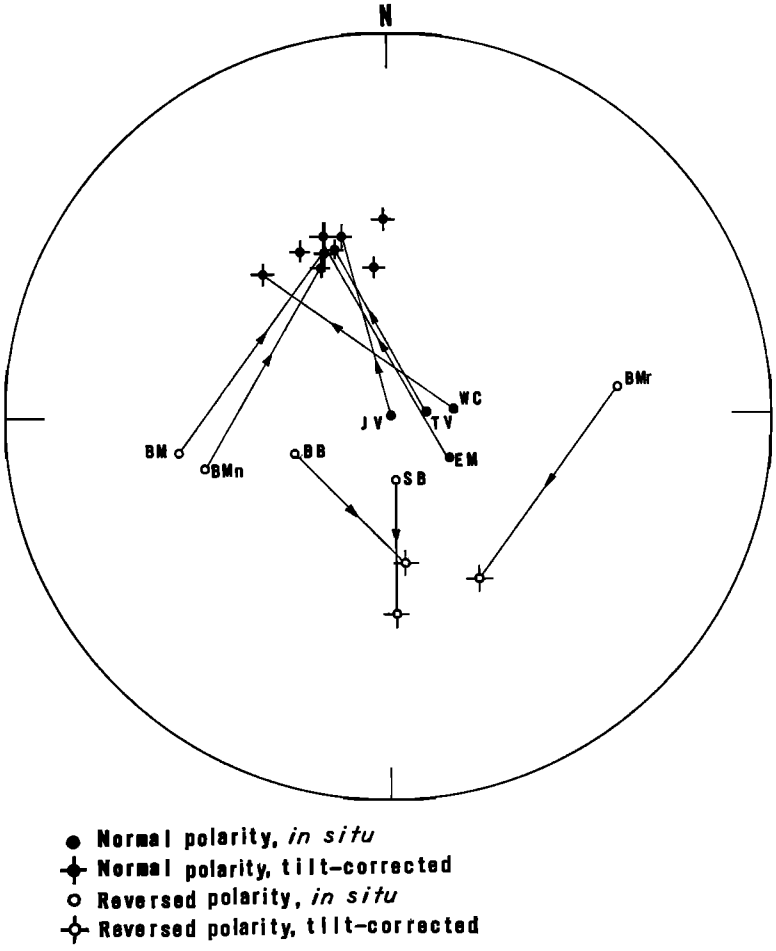


Fig. 7. Mean NRM directions of folded rocks before and after structural restoration. Tilt corrections were applied along the present strike, horizontal fold axes being assumed. The three reversed tilt-corrected directions are also shown in their antipodal lower hemisphere positions. Sites: Bm_n, Bull Mountain normal; BM_r, Bull Mountain reversed; BM, Bull Mountain normal and reversed; SB, South Boulder Valley; BB, Bone Basin; EM, southern Elkhorn Mountains; TV, Townsend Valley; WC, Whitetail Creek; JV, Jefferson Valley. Schmidt equal-area projection.

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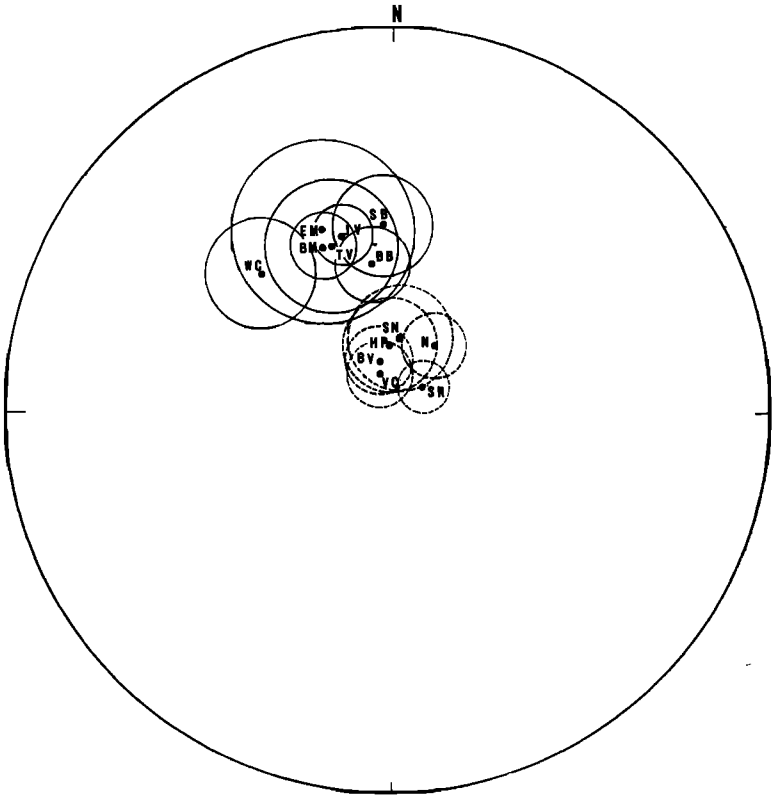


Fig. 8. Mean NRM directions of all folded and flat-lying rocks investigated with circles of confidence at the 95% probability level. Confidence circles associated with structurally restored folded rocks are unbroken; circles associated with flat-lying rocks are dashed. Sites of folded rocks: BM, Bull Mountain; SB, South Boulder Valley; BB, Bone Basin; EM, southern Elkhorn Mountains; TV, Townsend Valley; WC, Whitetail Creek; JV, Jefferson Valley. Sites of flat-lying rocks: S_n, Sappington normal; S_r, Sappington reversed; N, Norris; VC, Virginia City; BV, Beaverhead Valley; HP, Beaverhead Valley (Horse Prairie Creek). Lower hemisphere Schmidt equal-area projection.

The similarities in lithology and magnetic polarity between units at the Bone Basin and Bull Mountain sites suggest a correlation not previously realized. Such a correlation supports the belief of *Klepper and Smedes* [1959] that the Elkhorn Mountains Volcanics extend south of the Jefferson River. All units sampled at the Bone Basin site were used in determining the VGP. Bedding is fairly well defined.

5. *South Boulder Valley.* Six flows and one tuff were sampled from outcrops approximately 25 km south-southeast of the Bull Mountain site. Parts of five units have been remagnetized by lightning. Partial demagnetization results indicate that one flow is normally magnetized and that the other units are reversely magnetized. Although dispersion of directions within most units is great even after partial demagnetization, data of six units were used for determining the VGP. Bedding is very well defined.

6. *Jefferson Valley.* Six tuffs, three flows, two tuffaceous sandstones, and one intrusive were sampled near Lewis and Clark Cavern State Park, approximately 15 km southeast of the Bull Mountain site. Two units have been locally remagnetized by lightning, and samples of four other units possess soft magnetic components that may be viscous remanent magnetization (VRM). Petrologic and field evidence suggests that highly dispersed, cleaned NRM directions of two units are attributable to deposition of epiclastic materials at temperatures below the Curie points of the mag-

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netic mineral constituents. Partial demagnetization results indicate that all of the units sampled are normally magnetized. Mean directions of ten units were used for determining the VGP. Bedding is well defined.

Mean directions of all sites of folded rocks investigated are shown before and after structural restoration in Figure 7.

Sites of flat-lying rocks. As a possible check on the assumed Late Cretaceous age of the folded rocks investigated, NRM directions of nearby flat-lying volcanic rocks were compared with those of the folded rocks, as shown in Figure 8. The flat-lying extrusives and related intrusives have probable ages ranging from early Tertiary to early Quaternary. Four sites were investigated and the magnetic results are summarized below.

1. *Sappington.* One volcanic plug remnant and one mafic dike were sampled at exposures approximately 25 km southeast of the Bull Mountain site, near Sappington, Montana. Both intrusive bodies have been partially remagnetized by lightning. The age of the plug has been bracketed as post-Eocene and pre-Pleistocene [*Robinson,* 1963]; the age of the dike is post-Devonian and probably Late Cretaceous or younger. Partial demagnetization results indicate that the plug is normally magnetized and that the dike is reversely magnetized. Separate VGP's were calculated for the intrusive units.

2. *Norris.* Samples were collected from a basaltic flow remnant approximately 40 km southeast of the Bull Mountain site, near Norris, Montana. The flow is younger than late Tertiary basin fill deposits and is probably Pliocene or Pleistocene [*Tansley et al.,* 1933]. The uppermost parts of the flow have been remagnetized by lightning; partial demagnetization results indicate that the flow is reversely magnetized. The presence of ilmenohematite minerals suggests a possible self-reversed magnetization (S. Uyeda, personal communication, 1964); however, heating and cooling experiments have failed to reproduce an instantaneous self-reversal. Directions of partially demagnetized samples were used in the determination of the VGP.

3. *Virginia City.* Six flows and one tuffaceous siltstone were sampled approximately 65 km south of the Bull Mountain site, near Virginia City, Montana. The extrusive rocks here, like those at the Norris site, are believed to be Pliocene or Pleistocene [*Tansley et al.,* 1933]. Parts of three flows have been remagnetized by lightning. Partial demagnetization results indicate that all the units are normally

TABLE 2. Virtual Geomagnetic South Poles of Folded and Flat-Lying Rocks Investigated

Site	*N*	Longitude	Latitude	δp	δm	Probable Age
Bull Mountain, normal	6*	182°W	69°N	8	12	Late Cretaceous
Bull Mountain, reversed	7*	178°W	63°N	10	14	Late Cretaceous
Bull Mountain, normal and reversed	13*	171°W	69°N	7	10	Late Cretaceous
Southern Elkhorn Mountains	4*	162°W	67°N	18	27	Late Cretaceous
Townsend Valley	5*	163°W	70°N	14	21	Late Cretaceous
Whitetail Creek	5*	159°E	54°N	11	16	Late Cretaceous
Bone Basin	4*	147°W	81°N	9	12	Late Cretaceous
South Boulder Valley	6*	118°W	74°N	9	14	Late Cretaceous
Jefferson Valley	10*	156°W	71°N	6	8	Late Cretaceous or early Tertiary
Sappington, normal	5†	94°W	73°N	18	20	Post-Eocene, pre-Quaternary
Sappington, reversed	13†	87°W	53°N	7	7	Late Cretaceous or younger
Norris	12†	66°W	64°N	9	10	Pliocene or early Pleistocene
Virginia City	5*	119°W	60°N	13	13	Pliocene or early Pleistocene
Beaverhead Valley	4*	121°W	66°N	10	11	Eocene
Beaverhead Valley, Horse Prairie Creek	2*	109°W	72°N	17	18	Late Miocene or Pliocene

* Number of mean rock unit directions used in estimating VGP of mean site direction.

† Number of sample directions used in estimating VGP of mean site direction, where only one rock unit occurs at the site.

δp, δm, semiaxes of the 95% confidence oval associated with the VGP [*Cox and Doell,* 1960].

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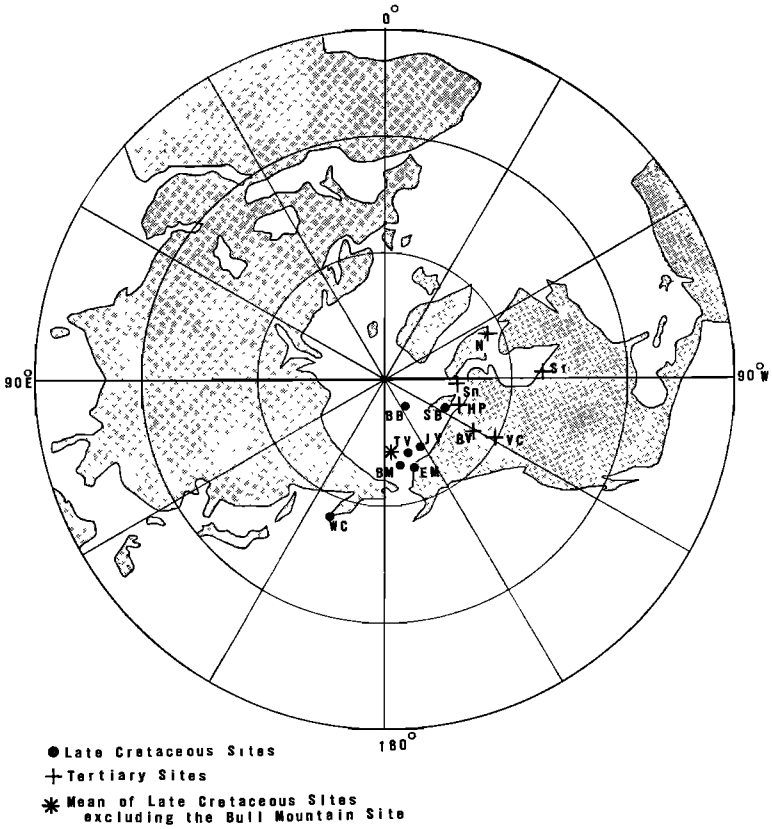


Fig. 9. Virtual geomagnetic south poles of all folded and flat-lying rocks investigated. Sites are designated as in caption of Figure 8. Schmidt equal-area projection of the northern hemisphere.

magnetized. Mean directions of five units were used for calculating the VGP.

4. *Beaverhead Valley.* Six flows, one flow breccia, one tuff, and one mafic intrusive were sampled in the Beaverhead River Valley and tributary Horse Prairie Creek Valley approximately 110 km southwest of the Bull Mountain site. Eight of the extrusive units in the Beaverhead Valley are believed to be Eocene (W. R. Lowell, personal communication, 1962), whereas one flow and one intrusive in the Horse Prairie Creek Valley are believed to be Miocene or Pliocene [*Lowell,* 1965]. Parts of four units have been remagnetized by lightning. Partial demagnetization results indicate that all units are normally magnetized. Mean directions of four units of probable Eocene age were used to determine one VGP; mean directions of the two later Tertiary units were used to determine a second VGP.

Summary and Discussion

The major part of the NRM in rock units investigated is primary, although secondary magnetic components have been added to outcrops of many rock units. Most of the secondary magnetization is believed to be isothermal remanent magnetization (IRM) produced by lightning; some secondary magnetization may be VRM acquired in the earth's field. Lightning-induced magnetization has been recognized by the localized occurrence of anomalous magnetization in isolated outcrops, by the general character of the outcrop terrain which in places includes lightning-struck natural objects such as trees, and by the typical hardness of the magnetization which may ordinarily be randomized in peak alternating fields of less than 450 oe.

Dispersion of NRM directions due to the

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addition of IRM or VRM is appreciably reduced after partial demagnetization. A few rock units, apparently mudflows, possess highly dispersed NRM directions before and after partial demagnetization, suggesting that their magnetic mineral constituents were randomly deposited at temperatures below their Curie temperatures. The possible contribution of pressure-induced magnetization (PRM) to the NRM of flows and intrusives has not been recognized; if there is a contribution, it is believed to be small. No evidence of inclination or bedding errors commonly associated with depositional remanent magnetization (DRM) was observed in the NRM of water-laid deposits; however, some or all of the NRM of these materials may be CRM. Although the possible effect of magnetic anisotropy on NRM directions is not known for those materials containing magnetic mineral constituents of high coercivity, the measured susceptibility anisotropy in weak magnetic fields cannot generally be related to NRM directions.

Mean site directions of all structurally restored folded rocks and all flat-lying rocks are shown in Figure 8, with their associated confidence circles at the 95% probability level. Although the confidence circles of all mean site directions are large, mainly because of the small number of data given equal weight, mean site directions may still be meaningfully compared with one another. For example, confidence circles associated with sites of folded rocks show considerable overlap, which suggests that the directions are not significantly different. In contrast, confidence circles associated with flat-lying rocks show very little overlap with circles associated with folded rocks. Although numerical comparison tests were not attempted owing to the variability of precision parameters, we may conclude that mean site directions of all folded rocks generally agree and that they differ significantly from mean site directions of flat-lying rocks.

VGP's corresponding to all folded and flat-lying rocks investigated are listed in Table 2 and plotted in Figure 9. Although ovals of confidence are not plotted, it may be noted that, of the seven ovals associated with folded rocks,

TABLE 3. Virtual Geomagnetic South Poles Reported for the Cretaceous of North America

Sampling Site and Reference	Longitude	Latitude	δp	δm	Remarks
Dakota Sandstone, Arizona [*Runcorn*, 1956]	160°W	75°N	9	11	Oval of confidence based on measurements of 10 specimens from 3 samples; sampling considered to be inadequate [*Irving*, 1964]
Boulder batholith, Montana (from data of *Colville* [1961])	167°E	76°N	6	7	Based on 15 samples of granodiorite cropping out near Butte
Mount Megantic intrusives, Quebec [*Larochelle*, 1961]	172°E	69°N	9	6	Based on 12 samples; radiometrically dated
Monteregian Hills intrusives, Quebec [*Larochelle*, 1961; 1962]	157°W	65°N	10	13	Based on 49 samples, 39 normal and 10 reversed; radiometrically dated
Isachsen Diabase, Northwest Territories [*Larochelle and Black*, 1963]	180°E	69°N	14	14	Based on 20 samples; radiometric date conflicts with age established on basis of stratigraphic and structural relations
Franciscan Formation, California [*Grommé and Gluskoter*, 1965]	50°W	26°N	12	18	Based on 7 site means of partially demagnetized rocks; paleomagnetic significance uncertain owing to structural complications (see reference)
Sierra Nevada granites, California [*Grommé and Merrill*, 1965]	165°W	69°N			Estimated mean of 7 VGP's; includes results of *Currie et al.* [1963]; radiometrically dated
Elkhorn Mountains Volcanics, Montana (this paper)	171°W	69°N	7	10	Based on 13 mean directions of rock units from Bull Mountain, 6 normal and 7 reversed

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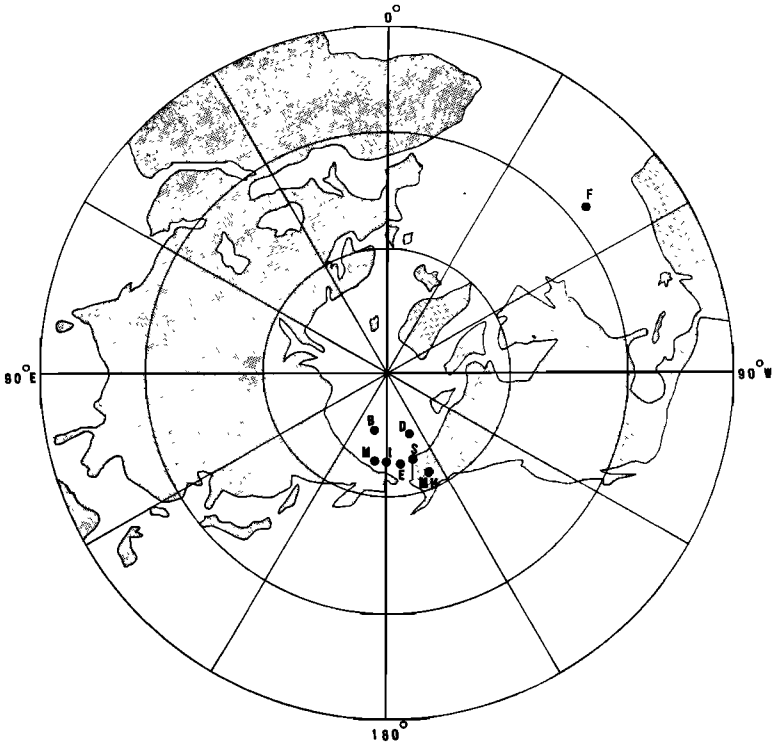


Fig. 10. Virtual geomagnetic south poles reported for Cretaceous rocks of North America. D, Dakota Sandstone, Arizona; B, Boulder batholith, Montana; M, Mt. Megantic intrusives, Quebec; MH, Monteregian Hills intrusives, Quebec; I, Isachsen Diabase, Northwest Territories; F, Franciscan Formation, California; S, Sierra Nevada granites, California; E, Elkhorn Mountains Volcanics, Montana. Schmidt equal-area projection of the northern hemisphere.

none appreciably overlaps either the present geomagnetic pole or the geographic pole. The VGP's associated with folded rocks are therefore significantly different from the present geomagnetic and geographic poles. Although all the folded rocks investigated are believed to be age-equivalent, only the age of rocks at the Bull Mountain site is considered to be definitely established. Figure 9 shows that the VGP corresponding to the Bull Mountain site lies within a few degrees of the mean of VGP's corresponding to all other folded rocks. The Bull Mountain pole is taken as the VGP of the Elkhorn Mountains Volcanics of Montana.

VGP's reported by various authors for the Cretaceous of North America are summarized in Table 3 and plotted in Figure 10 for comparison with the Bull Mountain pole (designated 'Elkhorn Mountains Volcanics, Montana'). The Bull Mountain pole lies within a few degrees of the mean of all poles calculated for the Upper Cretaceous Sierra Nevada granites of California [*Grommé and Merrill,* 1965] and within a few degrees of a pole calculated for the Isachsen diabase of the Northwest Territories [*Larochelle and Black,* 1963], although the Cretaceous age of the diabase is uncertain. The only North American Cretaceous pole considerably displaced from the Bull Mountain pole is that calculated for the Franciscan Formation of California [*Grommé and Gluskoter,* 1965], and the paleomagnetic significance of this pole is uncertain because of structural complications. The concurrence of the available North American Cretaceous data strengthens the assumption that the time-averaged ancient geomagnetic field approximated that of a geocentric dipole.

VGP's calculated by various authors for Cretaceous sites exclusive of North America

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are less tightly clustered than poles associated with North American sites. Poles listed by *Irving* [1964] for sites of Upper Cretaceous rocks from Madagascar, the U.S.S.R., and Argentina are not far removed from the three poles associated with Upper Cretaceous rocks in California and Montana. Poles corresponding to sites of Lower Cretaceous rocks from Japan and Australia are considerably displaced from poles associated with Lower Cretaceous rocks in North America. The divergence of the Japanese and Australian poles from the North American poles suggests possible Late Cretaceous movements of these land masses with respect to the North American continent. However, detailed interpretations of the ancient field configuration, apparent polar wandering, and relative movements of continents require considerably more data than are now available.

Conclusions

Paleomagnetic studies of volcanic rocks in southwestern Montana have established the following:

1. The main part of the NRM in the rocks is primary; that is, the magnetization of structurally tilted volcanic strata was acquired before folding.

2. Secondary magnetization components are mainly IRM produced by lightning, and these components may be effectively removed in peak alternating fields of 100 to 450 oe.

3. Reversed remanent magnetization occurs both in folded and in flat-lying rock units; the origin of this magnetization is tentatively inferred to be reversals of the ancient magnetic field.

4. A possible correlation of rock units based on similar petrologic properties and reversed magnetic polarities suggests that the field of the Elkhorn Mountains Volcanics extends south of the Jefferson River Valley, as proposed by *Klepper and Smedes* [1959].

5. The general agreement of structurally restored mean site directions for rocks believed to be age-equivalent on the basis of geological evidence suggests that the fold axes at the sites are nearly horizontal.

6. A part of the geological sequence of events accompanying the Laramide orogeny may be related to inferred magnetic field polarities. The relations are best interpreted at the Bull Mountain site, where the following sequence is inferred:

(*a*) Onset of regional folding and deposition of conglomerate—field polarity unknown.

(*b*) Deposition of tuffs and water-laid material with slight angular unconformity on older rocks—field polarity normal.

(*c*) Extrusion and deposition of flows, tuffs, and sediments before further folding—field polarity reversed.

(*d*) Emplacement of a sill before further folding, with the simultaneous baking of materials stratigraphically above and below the sill—field polarity normal.

(*e*) Completion of folding and emplacement of the Boulder batholith and related intrusives—field polarity normal (from data of *Colville* [1961]).

(*f*) Extrusion and intrusion of volcanic materials after major folding and batholithic intrusion—field polarities both normal and reversed, sequence unknown.

7. The virtual geomagnetic pole calculated for the structurally restored Upper Cretaceous rocks of the Bull Mountain site is 171°W, 69°N; this pole is not significantly different from poles calculated by other authors for the Cretaceous of North America.

8. The general agreement of VGP's reported for the Cretaceous of North America suggests that the ancient field may be approximated by that of a geocentric dipole.

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